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NMC 2016 A-LEVEL PAPER 1 QUESTIONS

SECTION A

Qn1. Suppose f is a function defined for all real numbers and

$$f(x) + 2f(3 - x) = x^2,$$

for all x. What is $f(1)$?

Qn2. The letters A, B, D and M represent the digits $1, 7, 8$ and 9, not necessarily in that order. Given that each of these digits is used exactly once, What is the largest possible sum of the three digit numbers BAD, DAM and MAD.

Qn3. Show that if $10a + b$ is a multiple of 7, then $a - 2b$ must also be a multiple of 7.

Qn4. In the sum below, each letter represents a different nonzero digit. If $R = 8$ and $F = 1$, determine the value of T and W.

$$TWO + TWO = FOUR$$

Qn5. Given that $\frac{a-b}{c-d} = 2$ and $\frac{a-c}{b-d} = 3$ for certain numbers a, b, c, d. Determine the value of $\frac{a-d}{b-c}$.

Qn6. Find the sum of all integers between 1 and 100 that leave remainder 2 upon division by 6.

Qn7. Ralph went to the store and bought 12 pairs of socks for a total of \$24. Some of the socks he bought cost \$1 a pair, some of the socks he bought cost \$3 a pair, and some of the socks he bought cost \$4 a pair. If he bought at least one pair of each type, how many pairs of \$1 socks did Ralph buy?

Qn8. Joshua writes the numbers $1, 2, 3, \ldots, 99, 100$. He marks out 1, skips the next number 2, marks out 3, and continues skipping and marking out the next number to the end of the list. Then he goes back to the start of his list, marks out the first remaining number 2, skips the next number 4, marks out 6, skips 8, marks out 10, and so on to the end. Joshua continues in this manner until only one number remains. What is that number?

Qn9. In the case of the Grand Hotel theft, inspector Derrick knows that the thief is either one or more of Cathy (room 23), Richards (room 27) and Miriam (room 36). He also knows that

- if Cathy or Richards is guilty of the theft, then Miriam is also guilty,
- Miriam and Richards are not both guilty,
- if Richards is not guilty, then so is Cathy.

What is the room number(s) of the thief or thieves?

Qn10. A boat travels around an equilateral triangle. The first $\frac{3}{4}$ of the distance requires 3.5 hours, while the last $\frac{3}{4}$ of the distance requires 4.5 hours. The middle third of the distance takes 10 minutes longer than the first third of the distance. How many hours did it take to travel around the triangle?

SECTION B

Qn11. Determine all pairs of real numbers a and x that satisfy the simultaneous equations $5x^3 + ax^2 + 8 = 0$ and $5x^3 + 8x^2 + a = 0$.

Qn12. Let b be a number such that $0 < b \leq 1$ and $b^{2016} - 2b + 1 = 0$. Determine all possible values of $1 + b + b^2 + \cdots + b^{2016}$.

Qn13. Three distinct integers a, b and c satisfy the following three conditions.
- $abc = 17955$,
- a, b and c form an arithmetic sequence in that order, and
- $(3a + b), (3b + c), (3c + a)$ form a geometric sequence in that order. what is the value of $a + b + c$?

Qn14. (a) Amina read 15 books, one at a time. The first book took her 1 day to read, the second book took her 2 days to read, the third book took her 3 days to read, and so on, with each book taking her 1 more day to read than the previous book. Amina finished the first book on a Monday, and the second on a Wednesday. On what day of the week did she finish her 15th book?

(b) There are 300 white boxes and n red boxes in storage. Each box contains the same number of soccer balls. The total number of soccer balls in all of the boxes is $n^2 + 290n - 2490$. Determine n.

Qn15. ABC is a triangle with $< B = 90°, < A = 30°$. Points P, Q and R are on $\overline{AB}, \overline{BC}$ and $\overline{CA}$ respectively. PQR is an equilateral triangle, $\overline{BC} = 4cm$ and Q is the midpoint of $\overline{BC}$. Determine $\overline{PR}$.

NMC 2016 A-LEVEL PAPER 1 SOLUTIONS

SECTION A

Qn1. Substitute $x = 1$ and $x = 2$ into the equation to get

$$f(1) + 2f(2) = 1 \quad \text{and} \quad f(2) + 2f(1) = 4.$$

solving these two equations simultaneously for $f(1)$ gives

$$\frac{1}{2} - \frac{1}{2}f(1) + 2f(1) = 4$$

$$\frac{3}{2}f(1) = \frac{7}{2}$$

$$f(1) = \frac{7}{3}.$$

Qn2. $BAD + DAM + MAD$ when expanded give us $(100B + 10A + D) + (100D + 10A + M) + (100M + 10A + D) = 102D + 101M + 100B + 30A$. Since we want the largest sum, we set $D = 9, M = 8, B = 7$ and $A = 1$ to get

$$(102 \times 9) + (101 \times 8) + (100 \times 7) + (30 \times 1) = 918 + 808 + 700 + 30 = 2456.$$

The largest possible sum of the three digit numbers is 2456.

Qn3. Since $10a + b$ is divisible by 7, then we write $10a + b = 7n$ where n is an integer. This means $b = 7n - 10a$. Substituting for b into $a - 2b$ we obtain $a - 2b = a - 2(7n - 10a) = a - 14n + 20a = 21a - 14n = 7(3a - 2n)$. Thus $a - 2b$ is a multiple of 7.

Qn4. $TWO + TWO = FOUR = 1OU8$. Clearly $2 \times O = x8$. If O is 4, then $x = 0$ or if O is 9, then $x = 1$.
Suppose that $O = 9$, then $TW9 + TW9 = 19U8$. In this case, $2T = 19$. Since $2T$ cannot be odd, it implies that 1 has been carried forward and added to 18. So $2T + 1 = 19$ so that $T = 9$ which is a contradiction.

In this case we conclude that $O = 4$ and we have $TW4 + TW4 = 14U8$. Clearly $2T = 14$ hence $T = 7$. Further more $2W$ can take on values $2, 4, 6, 8$. However 4 and 8 are excluded since $R = 8$ and $O = 4$. Also 2 is excluded since $F = 1$. Therefore we conclude that $W = 3$.

Qn5.

$$a - b = 2c - 2d \Longrightarrow a + 2d = b + 2c \tag{1}$$

$$a - c = 3b - 3d \Longrightarrow a + 3d = 3b + c \tag{2}$$

4eqn(3) − 3eqn(4) gives $a - d = -5b + 5c = -5(b - c)$.

Hence

$$\frac{a-d}{b-c} = -5.$$

Qn6. Let x be the integer between 1 and 100 such that $\frac{x}{6} = n\ rem\ 2$ where n is an integer. Then $x = 6n + 2;$ for $x = 0, 1, 2, 3, \cdots, 16$. Now

$$\sum_{n=0}^{16}(6n+2) = \sum_{n=0}^{16} 6n + \sum_{n=0}^{16} 2 = 6 \times \frac{16}{2} \times (16+1) + (2 \times 17) = 850.$$

Qn7. Let there be x pairs of \$1 socks, y pairs of \$3 socks and z pairs of \$4 socks. Then we have

$$x + y + z = 12 \tag{3}$$
$$x + 3y + 4z = 24 \tag{4}$$
$$x, y, z \geq 1 \tag{5}$$

Eqn(6)-Eqn(5) gives $2y + 3z = 12$.

It follows from the inequality 7 and the equation $2y + 3z = 12$ that y is a multiple of 3 and $2y$ is a multiple of 6. So since $0 < 2y < 12$, we must have that $2y = 6 \Longrightarrow y = 3$.

Since $y = 3$ then $z = \frac{12-6}{3} = 2$ and $x = 12 - y - z = 12 - 3 - 2 = 7$.
Thus Raphl bought 7 pairs of \$1 socks.

Qn8. Following the pattern of cancellation, we are crossing out;
Round 1: Every non-multiple of 2,
Round 2: Every non-multiple of 4,
Round 3: Every non-multiple of 8.
Following this pattern you are left with every multiple of 64 which is only 64 as the numbers are only 100.

Qn9. Suppose Richards is guilty of the theft then by the first statement Miriam is also guilty which contradicts the second statement. Thus Richards is not guilty. By the third statement, Cathy is also not guilty. Since at least one of the three persons is guilty, the only possibility is that Miriam is the thief.

Qn10. Let x, y, z denote the number of hours required to traverse each of the three sides. Then

$$x + y + \frac{z}{4} = \frac{7}{2}$$

$$\frac{x}{4} + y + z = \frac{9}{2}$$

$$y = x + \frac{1}{6}$$

Put the third equation into the first and second equations we get

$$\begin{aligned} x + x + \frac{1}{6} + \frac{z}{4} &= \frac{7}{2} \\ \frac{x}{4} + x + \frac{1}{6} + z &= \frac{9}{2} \end{aligned}$$

Hence we have

$$\begin{aligned} 2x + \frac{z}{4} &= \frac{20}{6} \\ \frac{5x}{4} + z &= \frac{26}{6} \end{aligned}$$

Solving these two equations simultaneously gives $x = \frac{4}{3}, z = \frac{8}{3}$ and $y = \frac{3}{2}$.

The total number of hours is

$$x + y + z = \frac{4}{3} + \frac{3}{2} + \frac{8}{3} = \frac{8 + 16 + 9}{6} = \frac{33}{6} = \frac{11}{2} \text{hours.}$$

SECTION B

Qn11. Subtract the two equations to get $ax^2 + 8 - 8x^2 - a = 0 \Longrightarrow (a-8)(x^2-1) = 0$. Thus $a = 8$ or $x = 1$ or $x = -1$.

Suppose $a = 8$ then $5x^3 + 8x^2 + 8 = (x+2)(5x^2 - 2x + 4) = 0$ and the second factor has no real roots since the discriminant $2^2 - 4(5)(4) = -76$ is negative. Hence $x = -2$ in this case.

If $x = -1$, we get $a = -5x^3 - 8x^2 = -5(-1)^3 - 8(-1)^2 = -3$.
If $x = 1$, we get $a = -5x^3 - 8x^2 = -5(1)^3 - 8(1)^2 = -13$.
Thus the three possible pairs are $(a, x) = (8, -2)$ or $(-3, -1)$ or $(-13, 1)$.

Alternatively

From $5x^3 + 8x^2 + a = 0$ we get $a = -5x^3 - 8x^2$. Thus

$$5x^3 + (-5x^3 - 8x^2)x^2 + 8 = -5x^5 - 8x^4 + 5x^3 + 8 = 0.$$

Factorising this polynomial equation
$-5x^5 - 8x^4 + 5x^3 + 8 = 5x^3(1-x^2) + 8(1-x^4) = 5x^3(1-x^2) + 8(1+x^2)(1-x^2) = (1-x^2)(5x^3 + 8x^2 + 8) = (1-x)(1+x)(x+2)(5x^2 - 2x + 4) = 0$.

Again we find that $x = -1, +1, -2$ since the quadratic factor $5x^2 - 2x + 4$ has no real roots.

Qn12. Let b be a number such that $0 < b \leq 1$ and $b^{2016} - 2b + 1 = 0$. Then

- Assume $b = 1$, then

$$1 + b + b^2 + \cdots + b^{2016} = 1 + 1 + 1 + \cdots + 1 = 2017.$$

• Suppose $0 < b < 1$, then

$$\frac{b^{2016}-1}{b-1} = 1 + b + b^2 + \cdots + b^{2016}.$$

From $b^{2016} - 2b + 1 = 0$ we have

$$b^{2016} - 1 + 2 - 2b = 0 \Longrightarrow b^{2016} - 1 = -2(1-b).$$

Substituting for $b^{2016} - 1$ into $\frac{b^{2016}-1}{b-1} = 1 + b + b^2 + \cdots + b^{2016}$, we get

$$\frac{-2(1-b)}{b-1} = \frac{2(b-1)}{b-1} = 2 = 1 + b + b^2 + \cdots + b^{2016}.$$

Hence

$$1 + b + b^2 + \cdots + b^{2016} = 2.$$

Qn13. a, b, c form an arithmetic progression in that order which means $a = b - d$ and $c = b + d$. We note that $d \neq 0$ and is the common difference. Writing the terms of the GP in terms of b and d we have

$$\begin{aligned} 3a + b &= 3(b-d) + b = 4b - 3d, \\ 3b + c &= 3b + (b+d) = 4b + d, \\ 3c + a &= 3(b+d) + (b-d) = 4b + 2d. \end{aligned}$$

Since $3a + b, 3b + c$ and $3c + a$ form a GP then

$$\frac{3b+c}{3a+b} = \frac{3c+a}{3b+c}$$

$$\begin{aligned} (3b+c)^2 &= (3c+a)(3a+b) \\ 12bd &= -7d^2, \end{aligned}$$

$$d = \frac{-12}{7}b$$

This implies $a = b - d = b - \frac{-12}{7}b = \frac{19}{7}b$ and $c = b + d = b + \frac{-12}{7}b = \frac{-5}{7}b$. Since $abc = 17955$, we have $\frac{-12}{7}b \times b \times \frac{-5}{7}b = 17955 \Longrightarrow b^3 = -9261 \Longrightarrow b = -21$. Thus $a = \frac{19}{7}b = -57$ and $c = \frac{-5}{7}b = 15$. We conclude that $a + b + c = -57 + 15 + -21 = -63$.

Qn14. (a) The process took $1 + 2 + 3 + \cdots + 14 + 15 = 120$ days so that the last day was 119 days after the first day. Now since 119 is divisible by 7, both must have been on the same day of the week. Hence Amina finished her 15th book on Monday.

(b) The number of balls in each box is

$$\frac{n^2 + 290n - 2490}{n + 300} = n - 10 + \frac{510}{n+300}$$

which is an integer. It follows that $\frac{510}{n+300}$ is an integer which is obviously positive and is less than $\frac{510}{300}$ i.e

it is less than 2. The only possibility is that $\frac{510}{n+300} = 1 \Longrightarrow n + 300 = 510$. Therefore $n = 210$. Hence there are 210 red boxes in the storage.

Qn15. Let $\overline{PB} = x, \overline{PQ} = y, \overline{BQ} = \overline{QC} = 2$. Since $< PQR = 60°$, it implies $< PQB + < RQC = 120°$ so that $< ACB = 60°$. Further more, $< RQC + < CRQ = 120°$ so that $< PQB =< CRQ$.

Let this angle be α and using the sine rule on $\triangle PBQ$, we have

$$\frac{PB}{\sin\alpha} = \frac{PQ}{\sin 90} \Longrightarrow \frac{x}{\sin\alpha} = y \Longrightarrow x = y\sin\alpha.$$

On $\triangle QCR$, we have

$$\frac{QC}{\sin\alpha} = \frac{RQ}{\sin 60} \Longrightarrow \frac{2}{\sin\alpha} \times \frac{y}{\frac{\sqrt{3}}{2}} \Longrightarrow y = \frac{\sqrt{3}}{\sin\alpha}.$$

Therefore

$$x = y\sin\alpha = \frac{\sqrt{3}}{\sin\alpha} \times \sin\alpha = \sqrt{3}.$$

We note by Pythagoras Theorem that

$$(\sqrt{3})^2 + 2^2 = (PQ)^2 \Longrightarrow PQ = \sqrt{3+4} = \sqrt{7} = PR.$$

Therefore $\overline{PR} = \sqrt{7}$.

NMC 2016 A-LEVEL PAPER 2 QUESTIONS

SECTION A

Qn1. Let a be a number such that $\log_a(10) + \log_a(10^2) + \log_a(10^3) + \ldots + \log_a(10^{10}) = 110$. What is the value of a?

Qn2. What is the smallest positive integer x for which $\frac{1}{32} = \frac{x}{10^y}$ for some positive integer y?

Qn3. A sequence is defined by $a_1 = 1$ and for $n \geq 1$ by;

$$a_{n+1} = \begin{cases} 0 & \text{if } a_n = 0 \text{ and } n \textit{ is} \text{ odd} \\ 2 & \text{if } a_n = 0 \text{ and } n \textit{ is} \text{ even} \\ 1 & \text{if } a_n = 1 \text{ and } n \textit{ is} \text{ odd} \\ 0 & \text{if } a_n = 1 \text{ and } n \textit{ is} \text{ even} \\ 1 & \text{if } a_n = 2 \end{cases}$$

How many of the numbers $a_1, a_2, a_3, \ldots, a_{2016}$ are equal to 2?